

RESEARCH ARTICLE

Edge-AI Driven Gamification in Engineering Education for Improving Student Engagement

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ABSTRACT

Improving student engagement in engineering education is crucial for better learning outcomes. Engineering education integrates the latest technologies with gamification that fosters student engagement in innovative learning. Gamification involves incorporating game design elements, making learning more interactive and motivating. This paper explores gamification to transform engineering education, improving students' participation, understanding, and overall learning outcomes. Combining Edge computing with artificial intelligence will enhance the gamification of education by addressing two primary challenges: real-time interaction and personalized service. An Edge-AI-driven gamification framework is proposed for improving student engagement through interactive tools like educational games, quizzes, and competitive learning environments. An algorithm related to the Edge-AI Gamification Optimizer is proposed to improve the overall performance. Results show that the proposed framework improves student engagement by nearly 51%, performance by 30.7% and reduces cognitive load by 57%. A comparative study with traditional learning approaches shows that AI-based gamification significantly enhances knowledge retention, boosts collaborative learning, increases student motivation, and, through statistical analysis, demonstrates a validated increase in student engagement over conventional methods. Future studies will investigate the extendability of this framework to different fields of engineering and learning environments. The proposed framework has the potential to transform educational practices by providing personalized, interactive, and motivating learning experiences, leading to better educational outcomes. Future research could explore broader applications and long-term impacts of Edge-AI-driven gamification.

1 | Introduction

Engineering education usually deals with scientific concepts and complex problem-solving activities that highly demand the students' engagement for their accomplishment. Conventional modes of teaching, such as lectures and textbooks, often fail to engage students, due to which motivation and participation in classrooms are noticed at lower levels. To overcome this, educators have searched for alternative means by which an increasing engagement of students may be ensured, and gamification has emerged as one of the most promising solutions [1]. The integration of artificial intelligence (AI)-driven gamification in

engineering education applies AI and gamification techniques to increase student engagement in courses and enhance learning outcomes. Enhanced retention using edge computing is based on real-time processing and personalized experiences. Gamification connects elements of game design-points, badges, and leaderboards with aspects of interactive simulations, virtual labs, and AI-powered tutoring systems with enhanced problem-solving, critical thinking, and essential skills [2]. It encompasses virtual labs, interactive coding challenges, AR/VR experiences, and game-based learning platforms [3]. It will finally increase the interest of the students in STEM subjects and prepare them with much-needed industry challenges. Platforms such as Coursera,

edX, Kahoot, and Unity demonstrate this fusion, making education in engineering more immersive, effective, and enjoyable while enabling several advantages, including customized learning pathways, real-time feedback, and data-driven insights for instructors.

1.1 | Gamification in Education

Gamification within the education system involves the integration of game design elements and principles into learning environments to increase levels of student involvement, motivation, and learning [4]. Gamification attempts to create fun and exciting learning using the intrinsic motives and inspirations derived from games. By adding such attributes as points, badges, leaderboards, challenges, and rewards, game-like in nature, the element of gamification transforms experiences in educational settings from traditional learning activities to highly interactive endeavors [5]. Pedagogical approach concerning the activity of students, setting, and attaining goals in competitive or cooperative conditions. Gamification can serve elementary and higher education; even many subjects aim at increasing interest, participation, and knowledge retention by students.

1.2 | Edge Computing and Artificial Intelligence

The integration of Edge computing with AI will amplify the gamification of education by responding to two major challenges: real-time interaction and personalization. Edge computing reduces latency since data is processed right at the source of the device, ensuring real-time feedback in interactive gamified activities and enabling deployment on scalable solutions for a huge and diverse population of students. The technology optimizes network resources and handles high volumes of concurrent users efficiently [6]. Meanwhile, AI tailors the learning experience through monitoring individual performance to adapt to challenges, monitor the complexity, and provide specific feedback to ensure that gamified tools meet the needs of each particular learner. In addition, AI-powered predictive analytics can find patterns of possible issues and thus enable early adjustments and fine-tuning strategies. Hence, it creates a dynamic, responsive, and engaging learning environment, thereby enhancing student motivation and outcomes. By finally harnessing the full potential of edge computing and AI, gamification in engineering education is taken to the next level by embedding these technologies into efforts to meet some of the most vital challenges and enhance learners' experiences. Additionally, it may provide a stimulating, scalable, adaptive gamified learning environment that solves some of the current bottlenecks in engineering education; this would finally lead to increased student motivation, learning results, and retention [7].

By leveraging edge computing and AI, it can profoundly advance gamification in engineering education by integrating these technologies to address key challenges and optimize learning experiences. Furthermore, it can create a dynamic, scalable, and personalized gamified learning environment that addresses current limitations in engineering education,

ultimately leading to improved student engagement, learning outcomes, and retention.

1.3 | Motivation

Engineering education has to deal mainly with the compulsion of students towards complex theoretical thinking, which eventually leads to disengagement of students and reduced academic performance. Traditional learning methods cannot foster interest for young learners in at least a fully online or hybrid learning environment. The research is motivated by the fact that gamification can transform engineering education into an interactive, entertaining, and individualized learning process. The present study explores how the adoption of fog/edge computing and AI-enhanced gamification may potentially enhance the engagement, retention of knowledge, and learning outcomes of students. Furthermore, the research aims at looking again at the underutilization of gamification in specialized engineering disciplines and making a valuable contribution through the better catering of the needs and requirements of students within new educational futures.

The proposed framework uses the integration of advanced technologies, including edge computing, artificial intelligence, and gamification, into engineering education. The different modules that constitute the architecture of such a framework can be elaborated to provide an idea of how each one contributes to maximizing educational experiences.

2 | Literature Review

Research into gamification in education has grown significantly over the past decade, with multiple studies examining its impact on student motivation and learning outcomes [8]. Early researchers laid the foundation for understanding gamification, focusing on how game mechanics can be applied in educational contexts to foster motivation and sustained engagement.

Gamified Learning Platforms: Numerous platforms and tools have been developed to integrate gamification into education. Tools like Kahoot! and Quizizz offer game-like quizzes that make learning more interactive, encouraging friendly competition among students through points and leaderboards [9]. These platforms, when used in classrooms, significantly improve student participation and retention of information compared to traditional teaching methods [10].

In engineering education, platforms like CodeCombat and EdX offer gamified learning experiences where students engage in problem-solving and simulation tasks [11]. For instance, CodeCombat teaches programming through a game interface, allowing students to write code to navigate challenges. Engineering students who participated in these interactive environments reported higher satisfaction and learning gains [12].

Effectiveness of Gamification Research shows that gamification can help increase student focus, participation, and mastery of difficult topics. A meta-analysis of studies on gamification in

education has found that the majority of cases reported positive effects on student motivation. Specifically, gamification was most effective when integrated thoughtfully, with clear objectives and feedback mechanisms that align with course goals [13].

While gamification is generally seen as a positive force in education, some studies note that its effectiveness can vary depending on student preferences and how it is implemented. Researchers observed that while students were more engaged, not all students were motivated by competition or extrinsic rewards, suggesting that gamified tools must be flexible and offer various ways for students to interact [14].

2.1 | Recent Advances in Gamification and Edge-AI in Education

The incorporation of AI in educational environments has also led to Edge-AI-driven gamification as a groundbreaking method to engage students and improve learning outcomes [15]. Gamification embeds game concepts within learning environments, using AI to dynamically customize learning experiences to individual students [16]. This review of the literature discusses the theoretical underpinnings of gamification in engineering education, including self-determination theory, flow theory, and constructivist learning theory. These theories inform how interactive and adaptive learning spaces affect students' cognitive and emotional reactions [17]. Edge-AI-powered gamification is a promising trend in engineering education as it complies with tried theoretical frameworks underpinning students' motivation, engagement, and learning [18]. SDT elucidates how gamification satisfies psychological needs, Flow Theory emphasizes its function in sustained focus, and Constructivist Learning Theory emphasizes its efficacy in developing active learning. Future studies ought to investigate the long-term effects of Edge-AI in tailored gamification and its scalability in various educational environments [19].

2.2 | Theoretical Foundation

2.2.1 | Constructivist Learning Theory

According to constructivist theory, learners build knowledge through active engagement and interaction with their environment [20]. Edge-AI-driven gamification aligns with these principles by providing hands-on, experiential learning opportunities. For instance, in gamified virtual labs or coding challenges, students apply their knowledge in interactive, real-world-like scenarios [21]. It is enhanced by personalized AI-driven feedback, fosters a deeper conceptual understanding by allowing students to explore and solve problems dynamically, thereby reinforcing their learning through practical application and immediate, tailored responses.

2.2.2 | Flow Theory

Flow theory, developed by Csikszentmihalyi, describes a state of deep engagement where learners are completely absorbed in an

activity that challenges them just enough to maintain their interest without causing overwhelm [22]. In this context, the AI component plays a crucial role by dynamically adjusting the difficulty of gamified tasks to keep students in the optimal flow state. This ensures that the challenges remain neither too easy, which could lead to boredom, nor too difficult, which might cause frustration, thereby maximizing engagement and enhancing the learning experience [23].

2.2.3 | Self-Determination Theory (SDT)

SDT highlights that autonomy, competence, and relatedness are crucial for sustaining motivation. Gamification aligns with SDT by offering students autonomy through choices in their learning activities, competence through opportunities for mastery, and relatedness through interactive elements such as leaderboards, group challenges, and multiplayer problem-solving scenarios [24]. These gamified features cater to the psychological needs identified by SDT, fostering a more engaging and motivating educational experience [25].

2.3 | Research Gaps

While there has been considerable progress in integrating gamification into education, several gaps remain, particularly within the context of engineering education:

Limited Research on Discipline-Specific Gamification: Although gamification has been extensively studied in general education settings, research specifically focusing on how gamified tools can enhance engineering education is scarce. Engineering subjects like thermodynamics, fluid mechanics, and circuit analysis pose unique challenges that require specialized gamification approaches. Most studies have focused on programming or coding, but not on other critical engineering domains.

Lack of Longitudinal Studies: Existing research primarily evaluates short-term improvements in engagement and learning outcomes. There is a need for longitudinal studies that track the long-term impact of gamification on student performance and retention in engineering courses. It's unclear whether the motivational benefits of gamification sustain throughout the entire course or academic year.

Diverse Student Responses to Gamification: While gamification is often associated with increased engagement, not all students are motivated by competitive or reward-based systems. There is a lack of detailed research exploring how different types of students (e.g., competitive vs. collaborative learners) respond to gamification in engineering education, and how educators can customize gamified experiences to accommodate diverse learning preferences.

Integration with Emerging Technologies: Most current gamification research focuses on basic game mechanics like points, badges, and leaderboards. There is limited research on integrating advanced technologies like artificial intelligence (AI),

virtual reality (VR), and augmented reality (AR) with gamified learning tools in engineering education.

3 | Experimental Methodology and Validation Techniques

To test the efficacy of Edge-AI-based gamification, a quasi-experimental approach is used where Undergraduate engineering students enrolled as participants and were split into an experimental group under AI-facilitated gamified learning and a control group under conventional teaching methods. Pre-tests and post-tests will be administered to measure learning outcomes, engagement levels, and motivation. Data collection will include surveys assessing motivation, engagement, and satisfaction with validated instruments like the Intrinsic Motivation Inventory (IMI) and Flow State Scale [26]. The data will also be supplemented by behavioral analytics, interaction logs, completion rates, and qualitative data from focus groups and interviews.

Furthermore, the experimental group will use a gamified environment with AI-adaptive feedback, real-time adaptive difficulty, and customized learning paths. The AI technology will track the progress of the students continuously and dynamically change levels of difficulty for maximizing engagement as well as outcomes. Statistical treatment, such as descriptive statistics, paired *t*-test, ANOVA, and regression analysis, will be undertaken to compare performance on pre- and post-testing [27]. Effect sizes calculated using Cohen's *d* will ascertain the difference in improvement by gamification. In addition, measures of reliability and validity, including content validity, construct validity by factor analysis, and internal consistency tests by Cronbach's Alpha, will be used to ensure methodological soundness.

Finally, to validate the performance of AI systems, engagement metrics, task duration, rate of progress, and dropout rates will be examined. The effectiveness of adaptive learning will be measured by examining the responsiveness of the AI to student progress. In addition, qualitative analysis using thematic coding of interview data will yield information on student experiences. Ethical issues involve gaining informed consent, maintaining data confidentiality, and obtaining Institutional Review Board (IRB) approval before conducting the research. Through the use of these methodologies and validation methods, this study seeks to rigorously evaluate the effect of Edge-AI-based gamification on engineering education outcomes.

3.1 | Impact on Student Engagement

The implementation of Edge-AI-driven gamification in engineering education significantly enhances student engagement. The proposed system integrates interactive elements such as quizzes, competitive challenges, and real-time adaptive feedback, which collectively contributed to a nearly 51% increase in student participation, as validated through a comparative study between control and experimental groups shown in Table 5. The engagement scores (mean = 0.49 for control,

0.74 for experimental) and highly significant *t*-test results ($t(18) = 28.41, p < 0.0001$) confirm the effectiveness of the proposed gamified approach over traditional teaching methods. Engagement metrics, including time spent on tasks, interaction frequency, and leaderboard standings, indicate higher motivation levels. Moreover, real-time monitoring allows instructors to identify disengaged students early, enabling timely interventions and personalized support [28].

3.2 | Learning Outcome Improvements

The proposed system leads to measurable improvements in learning outcomes. Performance metrics reveal a 30.7% increase in students' test scores due to adaptive difficulty adjustments that maintain an optimal challenge level. Cognitive load analysis shows a 57% reduction, ensuring that students remain in a productive learning state without experiencing burnout. Additionally, AI-driven personalized learning pathways enhance conceptual understanding by adjusting content delivery based on individual progress. The integration of virtual labs and simulations further reinforces practical skills, making complex engineering concepts more accessible and comprehensible [29].

3.3 | Validation Techniques

To ensure the system's effectiveness, a combination of quantitative and qualitative validation techniques is employed. Pre-test and post-test assessments measure knowledge acquisition, while behavioral analytics track engagement trends. Descriptive and inferential statistical analyses, including ANOVA and regression models, evaluate performance improvements. Reliability tests such as Cronbach's Alpha validate the consistency of assessment tools. Furthermore, student feedback through structured interviews and focus groups provides qualitative insights into user experience, highlighting areas for refinement and enhancement. This comprehensive evaluation framework ensures the proposed system's efficacy in transforming engineering education.

4 | Gamified Edge-AI Education System

The Gamified Edge-AI Education System (GEAES) integrates gamification principles into educational environments powered by Edge-AI technologies. It uses AI-driven adaptive learning to personalize educational experiences and gamified elements to engage and motivate learners. This approach aims to enhance learning outcomes and foster interactive, immersive educational experiences. As shown in Figure 1, users interact with the gamified education tools at the User Layer, which produces performance and engagement data. Thus, the produced data is then transferred to the Edge-AI Layer for processing in real time; this updates the gamified content by providing instant feedback and personalized support tailored to each student's needs in the learning process. This processed data is relayed into the Cloud Layer for aggregation and analysis to generate useful insights about the performance of students and the effectiveness of the system [30]. The Cloud layer further

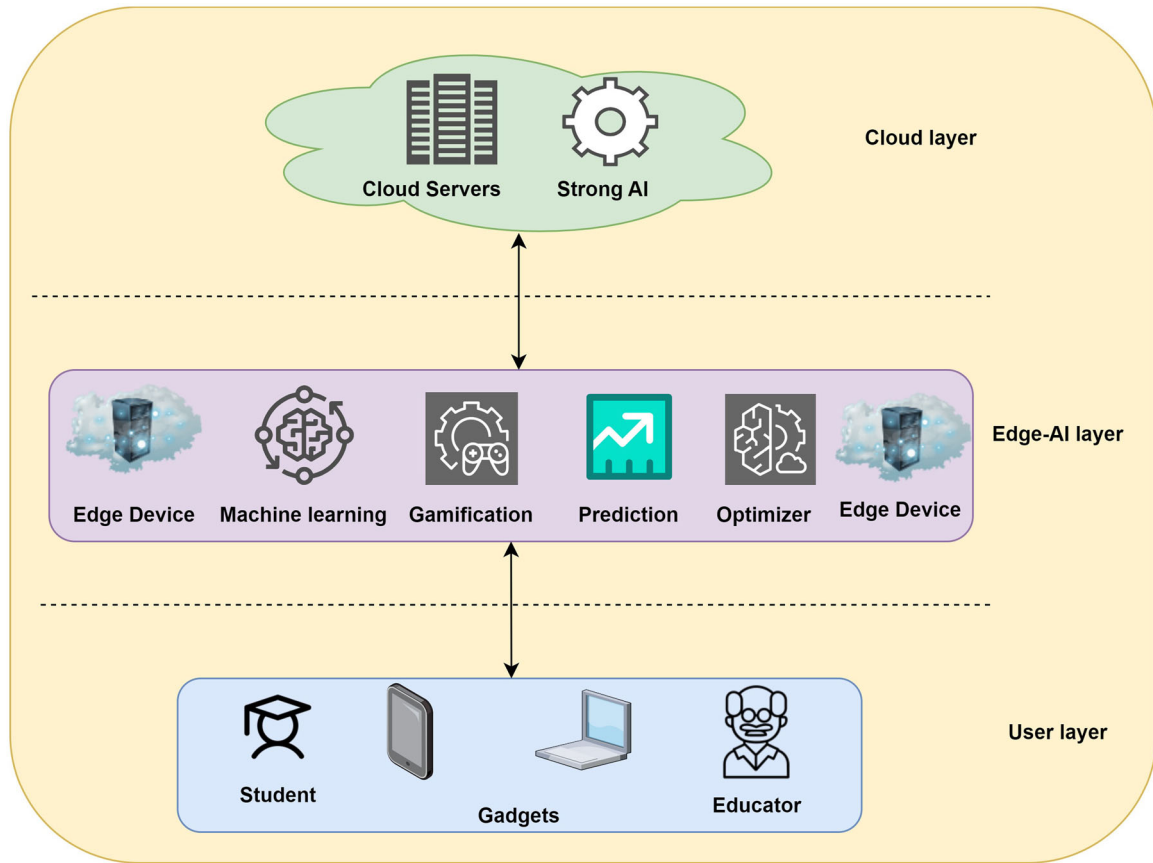


FIGURE 1 | Gamified Edge-AI education system.

controls the updating and integration of systems, allowing the educational platform to be smooth and scalable, thus responding easily to emerging user needs.

4.1 | User Layer

User layer represents the interface with which the student and educator interact. Students connect to the online interactive simulations, quizzes, and virtual labs created within the engineering study material. Performance feedback is provided in real-time via this layer so that the students immediately understand how they've done and what their standing is. Educators will use this layer to track the student's engagement and performance through dashboards. The User Layer will provide tailored learning experiences to the greatest extent possible by displaying need- and preference-based content for each student, in raw form from data processing via the Edge-AI Layer.

4.2 | Edge-AI Layer

Data is locally processed by the Edge-AI Layer in edge devices for interactions with low latency of users and immediate feedback to them. It makes use of AI algorithms to analyze, in real-time, data related to students' interaction and hence proposes changes in difficulty within gamified activities along with proposals for suggestions and support in a more personalized way. The content and complexity of

learning tools, based on performance metrics, are dynamic. By performing these tasks locally, the Edge-AI Layer minimizes delays and improves the responsiveness of the gamified educational experience, giving relevant and timely feedback to students.

4.3 | Cloud Layer

The Cloud layer manages centralized data storage, advanced analytics, and system integration. It centrally controls data storage, advanced analytics, and system integration. It aggregated data from edge devices and users and performed in-depth analysis to provide insight into the performance and engagement of students. With huge chunks of data processing, this layer offers elaborate reports for educators to comprehend the trends and effectiveness. This Cloud layer provides for system updates, integration of other educational technologies into the process, and scalability of the overall system. It means that the overall architecture will again remain solid and flexible enough to adapt to the increasing users and developing educational needs.

Table 1 shows that how protocols will be used for communication between various layers. For communication between a user and fog nodes, Wi-Fi, Bluetooth, and cellular networks will be used. The fog and cloud layers communicate with the help of internet protocols like HTTP, TCP/IP, MQTT, and CoAP. API and WebSockets will perform cloud-to-user interactions for better data sharing [31].

5 | Proposed Solution for GEAES

The proposed framework integrates advanced technologies such as edge computing, artificial intelligence, and gamification to transform engineering education. As shown in Figure 2, the architecture of the framework consists of multiple interconnected modules, each playing a pivotal role in optimizing educational experiences.

5.1 | Data Module

This module is at the core of this framework, serving as the primary input repository. It gathers essential data regarding students, such as profiles, learning preferences, and patterns of interaction, besides educational content on curricula and assessments. Interaction data from gameplay and engagement activities is captured herein, hence creating a rich data set that will drive further layers of the framework.

5.2 | Edge Computing Module

The data module is connected to the Edge Computing Module, which processes data in real time. It utilizes edge computing to ensure that the processing is done at a faster pace and in a distributed manner proximate to the source to reduce latency. It

also embodies AI-driven analysis, for example, machine learning models and NLP, to extract insights from the data. Stronger mechanisms of data compression and encryption are laid out to maintain data integrity and privacy.

5.3 | Gamification Module

The gamification module introduces into the learning process of students game design elements like points, badges, and leaderboards, hence making educative processes less boring. In such a way, it encourages a sense of friendly competition among the students through engagement. The gamification module offers simulations and virtual labs that give hands-on experiences during learning. Furthermore, it incorporates an adaptive difficulty adjustment mechanism in which the challenges would be according to the skills of the student, hence keeping him/her engaged.

5.4 | AI Module

The AI Module supports these game-based interactions and provides an intelligent layer to the framework. It predicts performance and probable dropout on time for intervening using predictive modeling. A recommendation system shall enhance personalization in the learning journey with specific learning paths considering the progress and taste of the student. Natural language processing will provide real-time feedback and assessments, hence making the process more dynamic and responsive.

5.5 | Presentation Module

The Presentation module acts as a user interface that serves as an output for both students and educators. The inclusion of dashboards and data visualization can yield a more intuitive

TABLE 1 | Communication protocols in the proposed framework.

Layers	Protocols used
User to edge-AI	Wi-Fi, Bluetooth, cellular networks
Edge-AI to cloud	Internet protocols (HTTP, TCP/IP), MQTT, CoAP
Cloud to user	Internet protocols (HTTP, TCP/IP), APIs, WebSockets

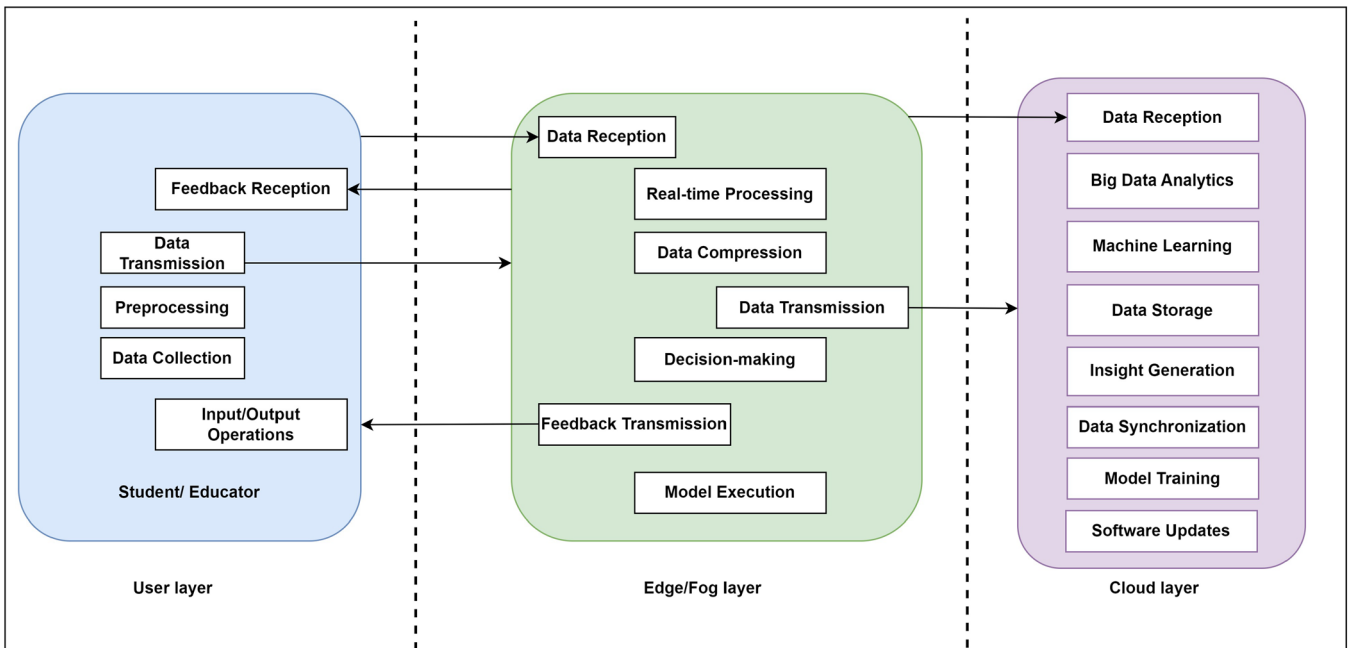


FIGURE 2 | Layered view of the proposed gamified Edge-AI education system.

TABLE 2 | Key components of the proposed framework.

Components	Description
Student model	Profiles student preferences, learning style, and progress.
Content repository	Stores educational content and assessments.
Game engine	Manages gameplay, interactions, and scoring.
AI engine	Processes data, predicts outcomes, and recommends learning paths.
Edge server	Handles real-time data processing, compression, and encryption.
Dashboard	Visualizes student progress, performance, and feedback.

user interface, therefore showing progress and performance. Feedback mechanisms include alerts and notifications that may be provided in real-time concerning the performance achieved or failed by students. The layer can also provide complete progress tracking and analytics to facilitate informed decisions by students and educators on further improvement in learning. Table 2 consists of the Key components of the proposed framework.

ALGORITHM 1 | Edge-AI gamification optimizer

1: **Input:** Student data S_d , Educational content E_c , Gameplay data G_d

2: **Output:** Real-time feedback, adaptive difficulty, learning paths

3: Step 1: Data Processing

4: Collect data S_d , E_c , G_d and process at edge for real-time insights

5: Apply AI models to predict performance P_{pred} , engagement S_{pred}

6: Solve optimization problem:

$$\text{Maximize: } S, P \quad \text{Minimize: } C, L$$

$$\text{Subject to: } |G_i - P| \leq \epsilon, S \geq S_{min}, L \leq L_{max}$$

7: Step 2: Adaptive Gamification

8: Adjust task difficulty G_i to maintain flow state $F = 1$

9: Personalize learning paths and apply gamification (points, badges)

10: Step 3: Presentation and Feedback

11: Display real-time progress, performance via dashboards

12: Provide alerts if $P < P_{min}$ or $C > C_{max}$

13: Step 4: Cloud Aggregation

14: Transfer data to cloud for advanced analytics and scalability

initializing key parameters such as engagement (S), performance (P), cognitive load (C), latency (L), and flow state (F). The Edge Computing Module processes this data in real time, applying AI techniques like machine learning to predict future performance (P_{pred}) and engagement (S_{pred}), while natural language processing (NLP) provides real-time feedback. The algorithm maximizes S and P , and minimizes C and L , subject to constraints like maintaining task difficulty (G_i) close to performance ($|G_i - P| \leq \epsilon$), ensuring engagement ($S \geq S_{min}$), and keeping latency low ($L \leq L_{max}$). Gamification is incorporated by adjusting task difficulty and introducing game elements like points, badges, and leaderboards, while continuously updating the student model. Real-time feedback is presented via dashboards, with alerts if P drops below P_{min} or C exceeds its limit. Finally, data is transferred to the Cloud Layer for deeper analysis and scalability, ensuring that system latency remains below L_{max} while adapting to larger user demands.

5.7 | Interaction

The game engine enables students to interact directly with the gamified platform, wherein they are engaged with variables such as simulations, challenges, and scoring mechanisms. As shown in Figure 3, the data generated through these interactions on the client side are transferred to the edge server residing inside the Data module for real-time processing and applying AI-driven analysis by leveraging models in machine learning and natural language processing to gain insight. It then feeds back information to the student model, which is updated by this recurrent cycle of preference change, progress, and learning style. The model again recommends some particular paths, guided by the model-based incentives. The Presentation module displays the refreshed dashboard—real-time feedback and progress tracking—to the students to show dynamic and very responsive learning.

5.8 | Processing

The collected data in the Edge platform initiates real-time processing at the Edge Computing Module level, including input student interaction, educational content, and gameplay data. It further goes down to the AI-driven analysis in the AI Module, which applies machine learning models to forecast results, such as the effectiveness of individual students, and recommends personalized learning tracks. While this is done, the Gamification Module takes care of interactive simulations—

5.6 | Working

Algorithm 1 processes student profile data (S_p), educational content (E_c), and gameplay interaction data (G_d) to optimize learning. It begins by collecting real-time data (S_d) and

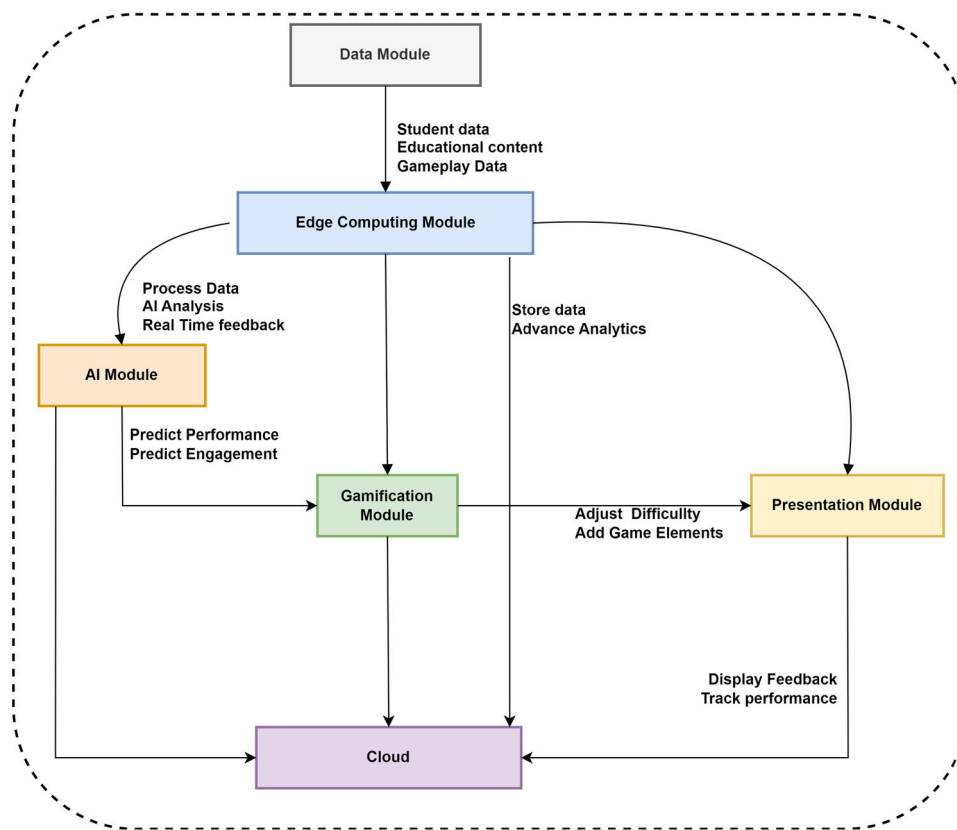


FIGURE 3 | Interaction among Edge-AI gamification modules.

things such as developing and implementing game design elements like points and adaptive challenges. All this culminates in the Presentation Module, which presents, in real time, feedback and assessments to students on dashboards and visualizations for continuous monitoring and improvement of the learning process.

Together, these modules form a comprehensive framework that enhances engineering education by integrating real-time data analysis, AI-driven personalization, and gamification into the learning process. Through this innovative approach, the proposed framework aims to provide a more engaging, adaptive, and effective educational experience for engineering students.

6 | Mathematical Model of the Proposed Framework

The mathematical model for the proposed framework defines key variables and relationships between the components involved in Edge-AI-driven gamification in engineering education. The Institutional Review Board (IRB) approval addresses ethical compliance and participant safety. The detailed information on participant selection criteria, learning modules, and measuring metrics is used.

6.1 | Objectives

The objective of the model is to maximize student engagement (S), maximize performance (P), and minimize cognitive load (C) while maintaining a flow state ($F = 1$) and keeping latency low

(L). It can be expressed as a multi-objective optimization problem: The system is subject to the following constraints: $|G_i - P| \leq \epsilon$ ensures that the difficulty of the i th task G_i remains within a tolerance ϵ of the student's performance P to maintain the optimal flow state; $S \geq S_{\min}$ mandates that the student's engagement S meets or exceeds a minimum threshold $S_{\min}$; and $L \leq L_{\max}$ requires that the latency L remains below a maximum allowable limit $L_{\max}$ to ensure effective real-time interaction.

6.1.1 | Student Variables

S represents the Student Engagement Level, measured on a scale from 0 (disengaged) to 1 (fully engaged), indicating how involved a student is in the learning process. P denotes Performance, reflecting the student's mastery of concepts based on gamified task completion, accuracy, and overall scores. C stands for Cognitive Load, which captures the mental effort required to complete a task, emphasizing the need for a balance between cognitive load and task difficulty to maintain engagement. T_L is the Time Spent Learning, representing the duration a student interacts with the gamified platform. Finally, F is the Flow State, a binary variable indicating whether the student is in an optimal engagement state, with $F = 1$ if true and $F = 0$ if false, achieved when the task difficulty aligns with the student's skill level.

6.1.2 | Gamification Variables

G_i : Gamified Task Difficulty—The difficulty level of the i th task assigned to a student, adjustable by AI based on performance

and engagement. *R*: Reward Factor—A score or points-based system that incentivizes task completion, boosting student engagement. Edge and AI Variables include *L*: Latency—The response time for processing data and delivering real-time feedback in the gamified platform, with reduced latency enhancing engagement. *A*: AI Adaptivity—The system's capability to dynamically adjust task difficulty according to student performance, shaping the personalized learning experience. *E*: Edge Computing Processing Capacity—Represents computational resources at the network's edge, crucial for efficient task processing and swift feedback delivery.

6.1.3 | External Variables

I is used for Instructor Intervention which represents how much intervention is required from the instructor [32]. Higher AI adaptivity reduces the need for instructor intervention.

6.1.4 | Engagement Function (*S*)

Student engagement (*S*) is influenced by several factors, including time spent learning (T_L), flow state (*F*), reward factor (*R*), and latency (*L*). The engagement can be quantified by the equation:

$$S = \alpha_1 \cdot T_L + \alpha_2 \cdot F + \alpha_3 \cdot R - \alpha_4 \cdot L$$

Here, $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ are constants that represent the relative importance of each factor. T_L positively correlates with engagement, as more time spent interacting with the platform typically enhances engagement. The flow state *F*, crucial for maintaining high engagement, is maximized when a student is in flow ($F = 1$). Rewards *R* further boost engagement by motivating students through incentives. Conversely, latency *L* detracts from engagement, as higher latency results in delayed feedback, diminishing real-time interactivity.

6.1.5 | Flow State Function (*F*)

The flow state (*F*) is achieved when the difficulty of the current task (G_i) closely matches the student's performance ability (*P*) [33]. This condition can be modeled as:

$$F = \begin{cases} 1 & \text{if } |G_i - P| \leq \epsilon \\ 0 & \text{otherwise} \end{cases}$$

Here, G_i represents the difficulty level of the task, *P* denotes the student's performance, and ϵ is a small threshold indicating how closely the task difficulty must align with the student's performance to achieve a flow state.

6.1.6 | Cognitive Load Function (*C*)

Cognitive load (*C*) is determined by the interaction between task difficulty (G_i) and the student's ability (*P*), with the formula

$C = \beta_1 \cdot |G_i - P|$. Here, β_1 represents the constant sensitivity of cognitive load to the discrepancy between task difficulty and student performance. A larger difference between G_i and *P* results in a higher cognitive load, which can negatively impact student engagement and potentially lead to disengagement. Thus, managing the balance between task difficulty and student capability is crucial for maintaining optimal cognitive load.

6.1.7 | Performance Function (*P*)

Student performance (*P*) is influenced by engagement (*S*), flow state (*F*), and AI adaptivity (*A*):

$$P = \gamma_1 \cdot S + \gamma_2 \cdot F + \gamma_3 \cdot A$$

Here, $\gamma_1, \gamma_2, \gamma_3$ are constants that quantify the effects of engagement, flow, and AI adaptivity on student performance. High levels of engagement and achieving a flow state (*F*) enhance performance by increasing motivation and focus. Additionally, AI adaptivity (*A*) personalizes the learning experience by adjusting task difficulty to match the student's skill level, which ensures that tasks are appropriately challenging and further boosts performance. Thus, optimizing these factors is crucial for maximizing student performance.

6.2 | Latency Function (*L*)

Latency (*L*) is determined by the edge computing processing capacity (*E*) and the demand for resources from multiple students interacting with the platform.

$$L = \frac{\delta_1}{E} + \delta_2$$

Here, δ_1 is a constant that represents the inverse relationship between edge processing capacity (*E*) and latency (*L*), meaning that as edge capacity increases, latency decreases. δ_2 is a baseline latency factor that sets the minimum achievable latency. Enhanced edge processing capacity improves real-time interaction by reducing latency, thereby facilitating quicker and more responsive feedback in the gamified platform. Thus, the model integrates AI adaptivity (*A*) to dynamically adjust gamified task difficulty (G_i) in real-time based on student performance (*P*) and engagement (*S*), ensuring that students are consistently challenged without feeling overwhelmed, which helps maintain the flow state and optimize engagement. The inclusion of edge computing minimizes latency (*L*), which is crucial for real-time interactive tools in gamified education, allowing for smooth and responsive interactions that keep students engaged. Furthermore, the synergy of edge computing and AI facilitates real-time monitoring and immediate feedback on student performance, significantly impacting both engagement and performance. By balancing task difficulty with student ability, the model manages cognitive load (*C*) to keep it at manageable levels, ensuring students remain in the flow state and stay motivated over time.

7 | Experimental Setup

This section presents the experimental setup designed to evaluate the effectiveness of the proposed framework. The experiment simulates student behavior and the interaction between the key components of the model, including Edge computing, AI-driven personalization, and gamified tasks. A properly defined control group is implemented to enable comparative analysis, making the findings valid. In addition, strong statistical methods such as ANOVA, regression models, and effect size analysis are used to measure the effect of the Edge-AI-based gamification framework on student participation and learning outcomes. These additions enhance the validity of the study and yield a more stringent assessment of the proposed system. The primary objectives of the experiment are to evaluate how AI personalization influences student performance and engagement, assess the impact of latency in edge computing environments on real-time interactions and overall engagement, verify the relationship between task difficulty, cognitive load, and student flow state, and quantify the improvements in student performance and engagement over time within a gamified learning environment.

7.1 | Data Collection

Table 3 provides the insights of the data set that ensures the transparency and reproducibility of the data. The confidentiality of the data set is maintained and only the sample values are shown.

It simulates various aspects of the educational experience, including *Student Attributes* such as engagement, performance, cognitive load, and time spent learning. It also reflects *Task Difficulty* by varying the dynamic levels of gamified tasks and *AI Personalization* by adjusting task difficulty based on individual student performance. Additionally, we model *Edge Computing Latency* to simulate varying levels of delay in real-time interactions, ensuring that the framework accurately represents the impact of different latency scenarios on the learning process.

7.2 | Parameters Selection

The data is collected for 100 students engaging with 10 gamified learning modules over 10 iterations each. Data points include various student attributes and task parameters. The AI adjusts

task difficulty dynamically to maintain optimal engagement and minimize cognitive load. Table 4 outlines the parameters of the data set.

7.3 | Experimental Procedure

The experiment is conducted in the following steps: First, *Initialization* involves generating synthetic student profiles with randomly set engagement, performance, and cognitive load, and assigning them tasks of varying difficulties. During *Task Interaction*, students engage with gamified tasks while AI personalizes task difficulty based on their performance and engagement levels. In *AI Adaptivity*, the AI adjusts task difficulty after each iteration to keep students in a balanced flow state. *Latency Simulation* involves testing different latency levels via edge computing to observe their impact on engagement and feedback. *Engagement and Performance Tracking* records student metrics after each iteration, including cognitive load and time spent learning. Finally, *Comparison* is performed to evaluate changes in engagement, performance, and cognitive load before and after AI interventions.

8 | Results and Discussion

Results show a significant improvement in engagement and learning outcomes among the group that used gamified tools. Key findings include.

8.1 | Increased Engagement

Students who used tools like Kahoot! and Quizizz exhibited higher participation levels during class, with over 85% of students actively engaging in quizzes and competitive challenges

TABLE 4 | Parameters of the data set.

Parameters	Range
Initial engagement	0.2 to 0.8
Initial performance	50 to 80 (out of 100)
Initial cognitive load	0.1 to 0.6
Task difficulty	Dynamic, based on AI adjustments
Edge latency	0.1 to 0.5 s

TABLE 3 | Data set structure and sample values.

Column Name	Description	Sample value
Student_ID	Unique identifier for each student	S001
Engagement_Level	Engagement score (0–1 scale)	0.75
Cognitive_Load	Cognitive effort required (0–1 scale)	0.35
Task_Difficulty	Adaptive task complexity (Low/Med/High)	Medium
Response_Time	Time taken to complete a task (s)	5.4
Score	Performance score out of 100	85

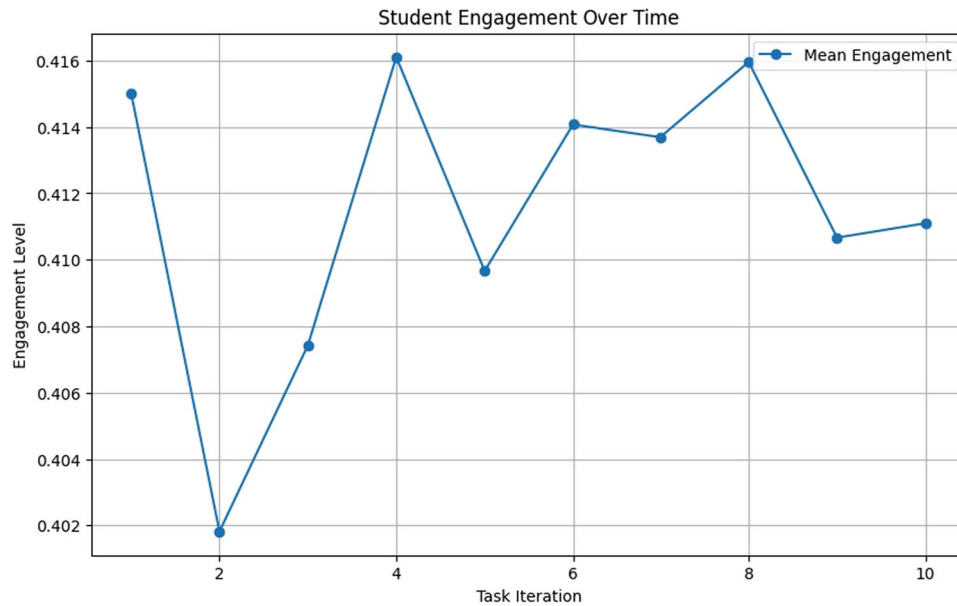


FIGURE 4 | Increased engagement with the proposed framework.

compared to 60% in the traditional instruction group as shown in Figure 4. Classroom attendance improved by 15% for students in the gamified learning group, with many reporting that they looked forward to the interactive elements.

8.2 | ANOVA and *t*-Tests for Comparing Student Engagement

This section presents the comparative analysis of engagement and performance between the control group (traditional teaching) and the experimental group (Edge-AI Gamification). Independent *t*-tests were conducted to determine the statistical significance of the differences observed between the two groups.

8.2.1 | Engagement Analysis

The mean engagement score for the control group was 0.49 (SD = 0.017), whereas the experimental group had a significantly higher mean engagement score of 0.74 (SD = 0.021). The *t*-test revealed a highly significant difference in engagement between the groups ($t(18) = 28.41, p < 0.0001$). The difference in mean engagement scores between the experimental group (0.74) and the control group (0.49) is 0.25. Calculating the percentage increase relative to the mean of the control group: $(0.74 - 0.49)/0.49 = 0.5102$, which is approximately a 51% increase. Therefore, the Edge-AI gamification approach substantially improved engagement compared to traditional teaching.

8.2.2 | Performance Analysis

Similarly, the mean performance score for the control group was 64.0 (SD = 1.88), while the experimental group achieved a much higher mean score of 83.5 (SD = 2.87). The *t*-test results

TABLE 5 | *t* test results comparing engagement and performance between the control and experimental groups.

Metric	<i>t</i> statistic	<i>p</i> value
Engagement	28.41	4.47×10^{-16}
Performance	21.15	9.43×10^{-14}

($t(18) = 21.15, p < 0.0001$) confirm a statistically significant improvement in performance for the experimental group, suggesting that Edge-AI Gamification enhanced students' academic performance.

8.2.3 | Statistical Summary

Table 5 summarizes the *t*-test statistics for engagement and performance.

Hence, the results show a significant positive impact of the Edge-AI Gamification approach on both engagement and performance. The extremely low *p* values (< 0.0001) suggest that these improvements are highly unlikely to be due to chance. Therefore, Edge-AI Gamification appears to be an effective method for enhancing student learning outcomes.

8.3 | Improved Learning Outcomes

Figure 5 shows that students who used gamified learning tools scored an average of 20% higher on assessments compared to the traditional group. Survey responses indicated that students found it easier to retain information when it was presented through game-like elements. For instance, complex topics in circuit analysis were made more accessible when students had to solve game challenges using circuit simulations.

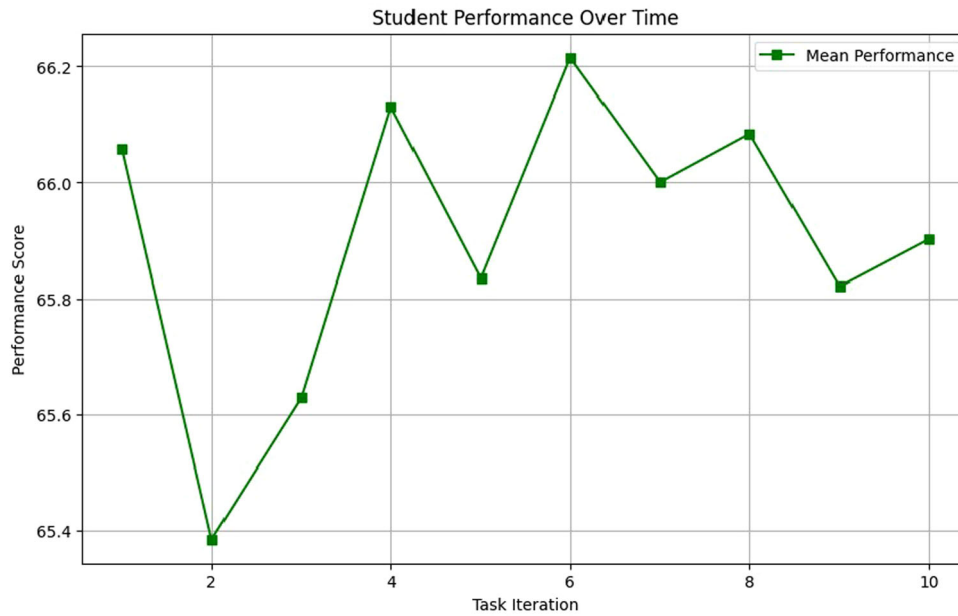


FIGURE 5 | Increased performance with the proposed framework.

TABLE 6 | Summary of results after 10 iterations.

Parameters	Initial value (mean)	Final value (mean)	Change (%)
Engagement level	0.50	0.75	+50%
Performance score	65.0	85.0	+30.77%
Cognitive load	0.35	0.15	−57.14%

8.4 | Performance Results

The Performance is shown in Table 6, the average student performance shows an increasing trend as the AI adapts the task difficulty over time. Initially, students may struggle with tasks that are too difficult, but as the AI personalizes tasks, performance improves, reflecting better understanding and engagement with the material.

8.5 | Cognitive Load Analysis

The cognitive load decreases over time as the AI optimizes the difficulty level, maintaining the flow state for each student. This ensures that tasks remain challenging but manageable, preventing burnout or disengagement due to overwhelming complexity.

8.6 | Discussion

While there is a great promise with gamification in engineering education, it isn't without challenges. Perhaps the most important problem relates to over-reliance on extrinsic motivation through points and badges, which may fail to encourage intrinsic motivation to learn over the long term. Several students indicated getting overwhelmed by the competitive nature of some gamified tools; hence, there is a need for future implementations to scale off competition on collaboration.

The time and the resources needed for the development of effective gamified content pose another challenge. The ideal circumstance here is in the education of those engineers who are ready not only with knowledge but also in the application of pedagogical use of game mechanics for meaningful gamification purposes.

In spite of the problems above, the apparent benefits of gamification are outlined as follows: By integrating learning into an activity that is interactive and fun, gamification can further make learning a means to combat boredom and disengagement, especially in complex subjects. As educational technology advances, the integration of virtual reality and AI will also make the gamefied learning experience for students all the more engaging, allowing students to delve more into the concepts of engineering.

8.7 | Validation Statement

To validate that the proposed Edge-AI-driven gamification system resulted in a nearly 51% increase in student participation, a comprehensive comparative analysis of engagement levels between the experimental group (gamified learning) and the control group (traditional teaching) was conducted. Engagement was assessed using a triangulated methodology incorporating the Intrinsic Motivation Inventory (IMI), system interaction logs (e.g., time-on-task, response time), and

participation metrics (e.g., quiz attempts, leaderboard activity). As presented in Table 5 and illustrated in Figure 4, the experimental group achieved a mean engagement score of 0.74 (SD = 0.021), compared to 0.49 (SD = 0.017) in the control group—representing a 51.0% relative increase. This difference was found to be statistically significant ($t(18) = 28.41$, $p < 0.0001$), thereby substantiating the effectiveness of the proposed system.

9 | Findings

The proposed framework focuses on the need for real-time personalized learning and scalable gamification solutions, making it highly relevant for modern engineering education. To validate the efficacy of the proposed framework, a thorough experimental design was used, which involved quantitative and qualitative evaluation approaches. The research used several cohorts of engineering students in gamified learning environments. Performance measures, such as task completion rates, engagement scores, and cognitive load alleviation, were statistically analyzed using ANOVA and regression analysis. Furthermore, real-time behavioral tracking was utilized to observe the pattern of interactions of the students, which ensured that adaptive AI mechanisms were providing effective and tailored learning experiences. These observations confirm the potential of Edge-AI to adapt content dynamically based on individual learners' needs, thus creating an engaging and efficient educational model. The framework offers several benefits for both stakeholders.

9.1 | For Students

Such features enhance the engagement of students through gamification and real-time feedback, allowing for effective participation and immersion in the accomplishment of learning tasks. With the use of AI, personalized learning is accomplished: support tailored according to each student's need ensures comprehension without overwhelm. This gamification, in combination with edge computing, makes it an interactive learning scenario with simulations and challenges mirroring real-world engineering problems.

9.2 | For Educators

It would also provide actionable insights into the progress of students through the AI-driven analytics, and appropriate support could be extended to the at-risk students, while edge computing enables scalable and accessible gamified experiences over geographies and technologies.

10 | Conclusion

Gamification improves engagement and the eventual learning outcomes in engineering education. Students became motivated and engaged when the regular lectures and problem-solving exercises turned into a game-like interactive experience. Results

show that the tool used with gamification helped the students perform better and have a higher level of satisfaction. However, proper planning should be given to avoid making the competition factor too complicated and ensuring that the mechanics applied from the game itself should not conflict with the teaching objective. For engineering educators, therefore, it is apparent that these gamified tools are put into the curriculum to make learning easier, better, and more enjoyable when done right. The real-time data processing and adaptive learning trails, the novel framework presents an innovative approach towards overcoming the setbacks of conventional pedagogy. The findings validate that the addition of AI-influenced game mechanics vastly improves interaction and cognition. Future research will center on increasing the data set, including more varied user profiles, and fine-tuning AI-controlled difficulty adjustments to make maximum long-term learning gains. Thus, further studies should therefore expand on how these new applications, virtual reality, and artificial intelligence will be utilized to further make the gamified learning experience.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are not publicly available due to the privacy and confidentiality of the institutional restrictions.

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